Task-Driven Hardware Optimization of CIS and DVS Image Sensor Architectures

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Introduction

Modern vision systems rely on algorithm–hardware co-design rather than pixel-level improvements alone. Sensor parameters such as frame rate, resolution, ADC precision, and event thresholds strongly affect power, latency, and task performance.

This project builds on ModuCIS (frame-based CMOS image sensor modeling) and a DVS power model. While these tools support basic parameter sweeps, they do not provide a task-driven method to identify optimal hardware configurations. Our project extends this baseline by systematically finding the best CIS and DVS settings for a chosen vision task under performance and power-efficiency objectives.

Aspect to Be Extended

The background material highlights limited algorithm-to-sensor coupling and lack of automatic optimization. We extend this work by introducing a **task-driven design-space exploration** framework that determines optimal CIS and DVS configurations under two goals:

1. Best task performance (allowing higher power)
2. Best power efficiency while maintaining acceptable task quality

Proposed Approach and Methods

Selected Vision Task

[Insert chosen task, e.g., motion detection]

A fair comparison setup will be used with identical assumptions (supply voltage, latency budget, and task conditions).

CIS Optimization (ModuCIS)

We will sweep:

- Frame rate
- Resolution
- ADC resolution

For each configuration, we will record power, latency, and task-quality proxy.

DVS Optimization

We will sweep:

- Event threshold
- Filtering policy
- False event assumptions

For each configuration, we will measure event rate, static/dynamic power, and quality proxy.

For both CIS and DVS, we will identify:

- **Performance-first configuration**
- **Power-efficient configuration**

Results will be summarized using Power vs Performance plots and tables.

Tasks Involved

1. Study baseline CIS and DVS models
2. Reproduce reference outputs
3. Define task metrics
4. Run CIS parameter sweeps
5. Run DVS parameter sweeps
6. Analyze power–performance tradeoffs
7. Generate plots and tables
8. Prepare report and presentation

Work Distribution

Member	Responsibility
Member 1	Task definition and baseline setup
Member 2	CIS parameter sweeps
Member 3	DVS modeling and sweeps
Member 4	Visualization, optimization summary, report

All members will contribute to final analysis and conclusions.

Expected Outcome

The project will produce optimal CIS and DVS configurations for both performance-first and power-first goals, along with clear design insights indicating when each architecture is preferable.

Conclusion

This project introduces task-driven hardware optimization for CIS and DVS image sensors. By systematically sweeping design parameters and identifying optimal operating points, we bridge algorithm requirements and sensor behavior, providing practical guidance for low-power vision system design.